Course Name: Basic Electrical and Electronics Engineering Course code: BSC 102

FEAIML (A+B)

1. Define RMS and Average value of an alternating current. Define amplitude, cycle, period, instantaneous value, frequency, for single phase AC supply with suitable waveform
2. Derive the formula for resonant frequency of a series resonant circuit. State any five conditions of series resonance .Compare series and parallel resonance
3. A resistor of 20 Ω, inductor of 0.1 H and a capacitor of 50 μF are connected in series. A supply voltage 230 V, 50 Hz is connected across the series combination. Calculate the following: (i) impedance, (ii) current drawn by the circuit, (iii) phase difference and power factor, and (iv)active and reactive power and apparent power consumed by the circuit.
4. Three similar coils each having a resistance of 5 ohm and an inductance of 0.02 H are connected in star to a 440V, 50Hz, 3-phase supply. Calculate (i) line current and (ii) active and reactive power.(iii) Phase current (iv) Phase voltage (v) Power factor.
5. Two impedances of 12v + j16 ohm and 10 - j20 are connected in parallel across 230V, 50Hz, single phase ac supply. Find admittance, current and power factor of each branch
6. A circuit consists of a pure resistor and coil in series. Power dissipated in the resistor and in the coil are 1000 W and 250 W respectively. The voltage drops across the resistor and the coil are 200 V and 300 V respectively. Determine (i) value of pure resistance, (ii) resistance and reactance of the coil, (iii) combined resistance of the circuit, (iv) combined impedance, and (v) supply voltage
7. An inductive coil having inductance of 0.04H and resistance 25 ohms has been connected in series with another inductive coil of 0.2H and resistance 15 ohms. The whole circuit is powered with 230 v, 50 Hz mains. Calculate the power dissipation in each coil and total power factor
8. Determine the relationship between phase and line voltage and current for star connected balanced load across a three-phase balanced system.
   Determine the relationship between phase and line voltage and current for delta connected balanced load across a three-phase balanced system.

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